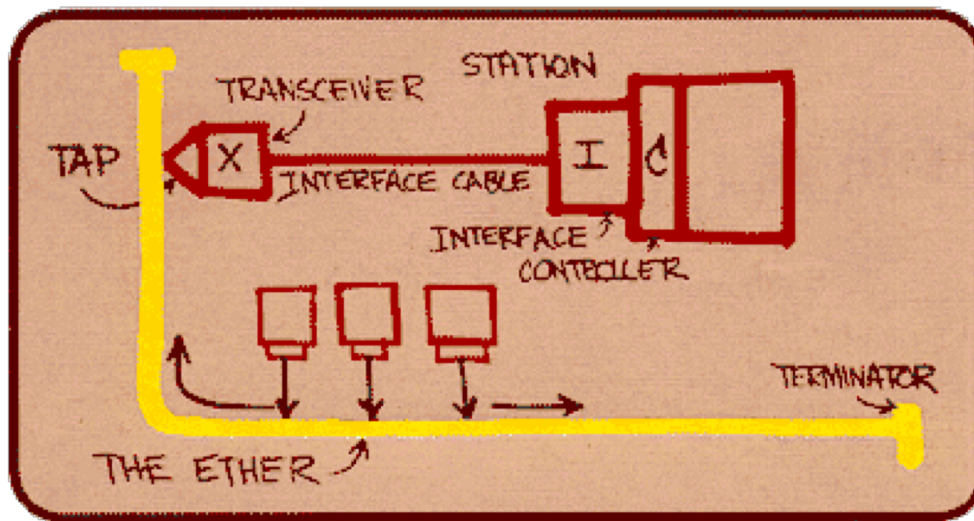


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“dominant” wired LAN technology:

- single chip, multiple speeds
- first widely used LAN technology
- simpler, cheap
- kept up with speed race: 10 Mbps – 10 Gbps

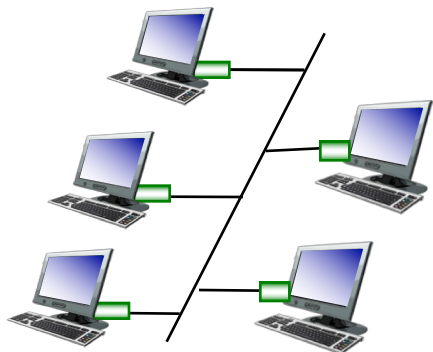


Metcalfe's Ethernet sketch

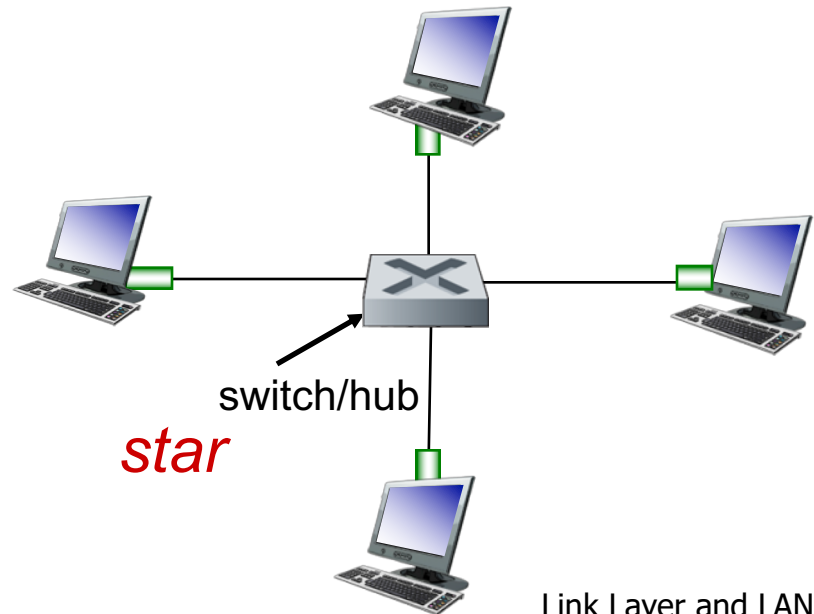
Ethernet: physical topology

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- **bus:** popular through mid 90s
 - all nodes in same collision domain (can collide with each other)
- **star:** prevails today
 - active **switch/hub** in center
 - each “spoke” runs a (separate) Ethernet protocol
 - switch: nodes do not collide with each other, buffers used
 - hub: all spokes broadcast their transmissions, no buffering



bus: coaxial cable



Ethernet frame structure

sending adapter encapsulates IP datagram (or other network layer protocol packet) in **Ethernet frame**



preamble:

- 7 bytes with pattern 10101010 followed by one byte with pattern 10101011
- used to synchronize receiver, sender clock rates

Ethernet frame structure (more)

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- **addresses:** 6 byte (octet) source and destination addresses
 - if adapter receives frame with matching destination address, or with broadcast address, it passes data in frame to higher layer protocol
 - otherwise, adapter discards frame
- **type:** indicates which higher layer protocol carried in frame
- **CRC:** cyclic redundancy check bits used by receiver to detect error
 - error detected: frame is dropped, no recovery

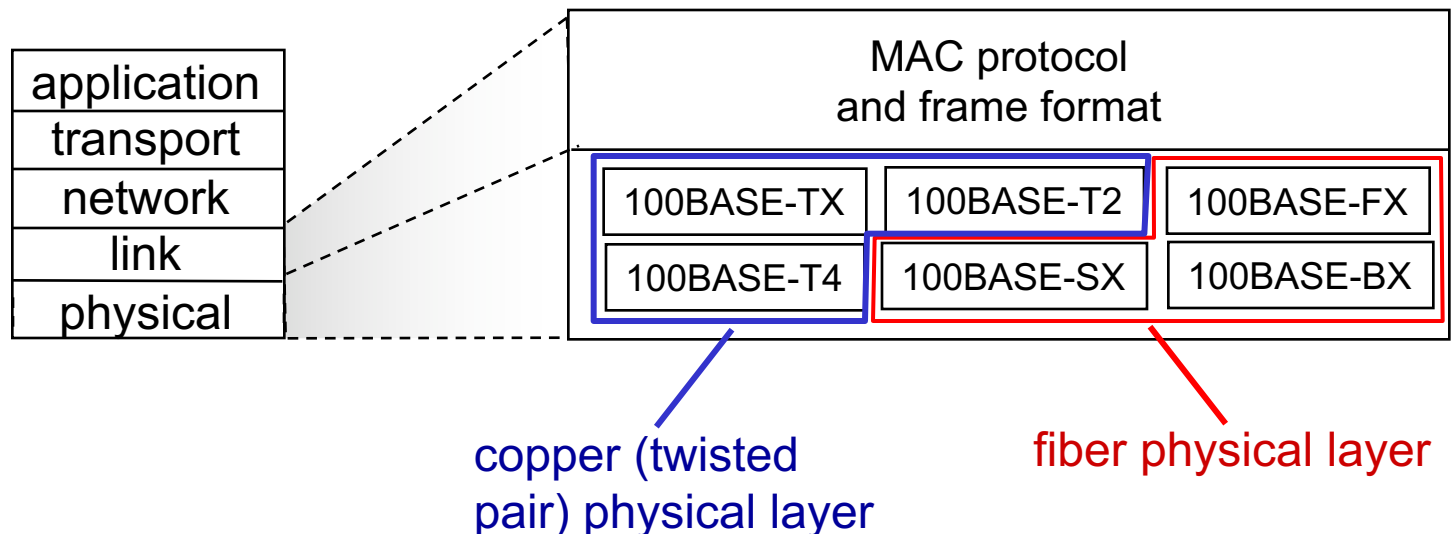


Ethernet: unreliable, connectionless

- *connectionless*: no handshaking between sending and receiving NICs
- *unreliable*: receiving NIC doesn't send acks or nacks to sending NIC
 - data in dropped frames not recovered
 - relies on higher layer reliable data transport (e.g., TCP), otherwise dropped data lost
- Ethernet's MAC protocol: unslotted *CSMA/CD with binary exponential backoff*

29 of 46 802.3 Ethernet standards: link & physical layers

- *many* different Ethernet standards
 - common MAC protocol and frame format
 - different speeds: 2 Mbps, 10 Mbps, 100 Mbps, 1 Gbps, 10 Gbps, 40 Gbps
 - different physical layer media: fiber, cable



Link layer, LANs: outline

6.1 introduction, services

6.2 error detection,
correction

6.3 multiple access
protocols

6.4 LANs

- Ethernet
- addressing, ARP

Device interface addresses

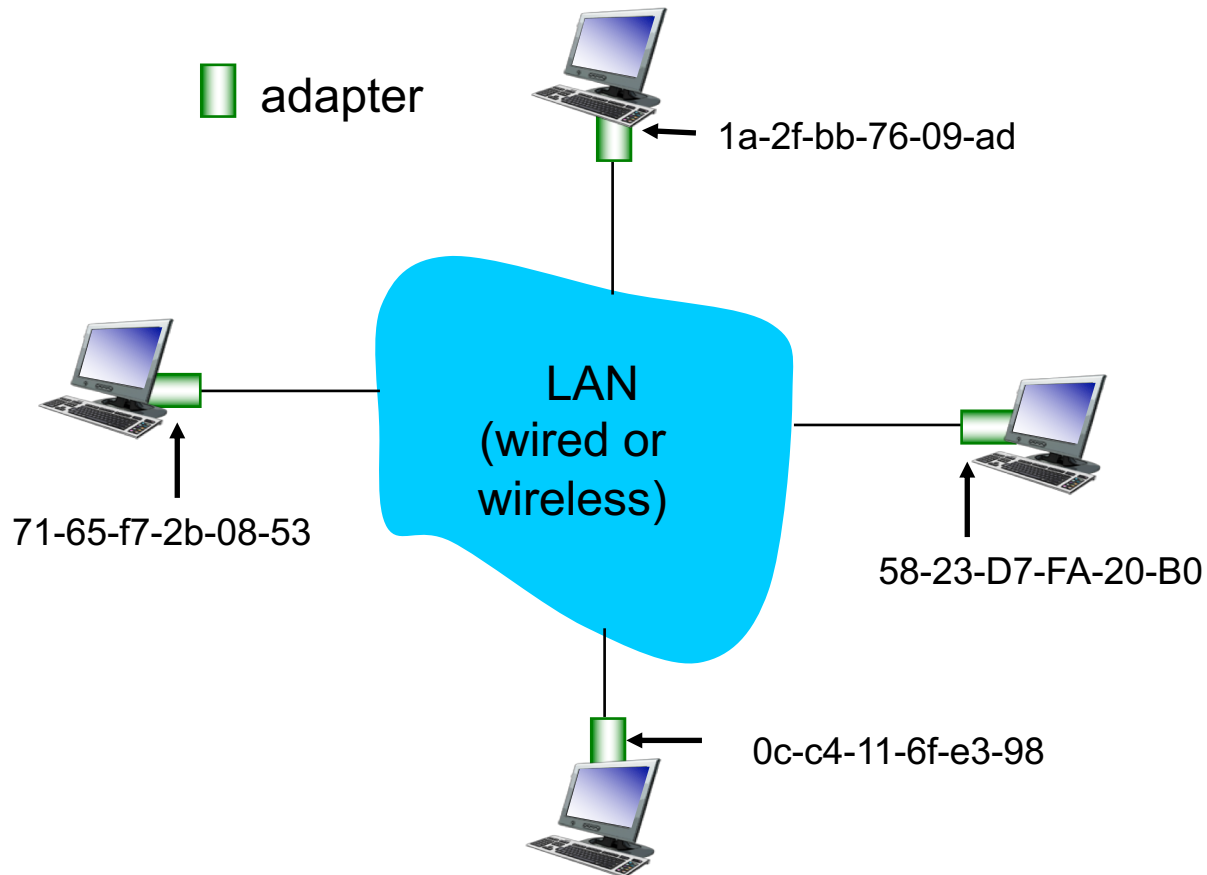
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- interface address (referred to as: MAC/LAN/Ethernet/physical/hardware address):
 - function: *used 'locally' to get data frame from one device interface to another device interface on same physical layer network – Ethernet, WiFi,.....,*
 - address bits burned in NIC ROM, also sometimes software settable
 - e.g.: Ethernet – 48bits (6octets): 1a-2f-bb-76-09-ad
 - hexadecimal (base 16) notation (each “numeral” represents 4 bits – $12 \times 4 = 48$)
 - 1=0001, 9=1001, 10=a=1010
11=b=1011, 14=e=1110, 15=f=1111
 - ff:ff:ff:ff:ff:ff – all “1”s -> **broadcast** address on local network

MAC addresses

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- each adapter on LAN has unique **LAN** MAC/hardware address
- devices can have more than one interface/adapter → one MAC address per adapter (Ethernet, WiFi, cellular)



MAC addresses (more)

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- MAC address allocation administered by IEEE
- manufacturer buys portion of MAC address space (to assure uniqueness)
- analogy:
 - MAC address: like **Social Security Number** (SSN)
 - Internet address: like **postal address**
- MAC -> flat address → **portability**
 - can move LAN card from one LAN to another
 - people move - their SSN goes along, does not change
- Internet -> hierarchical address (network-host) → **not portable**
 - address based on which **network** the device is currently connected to, e.g., UCI, Cox OC, Sprint, Starbucks
 - people move - their postal address changes

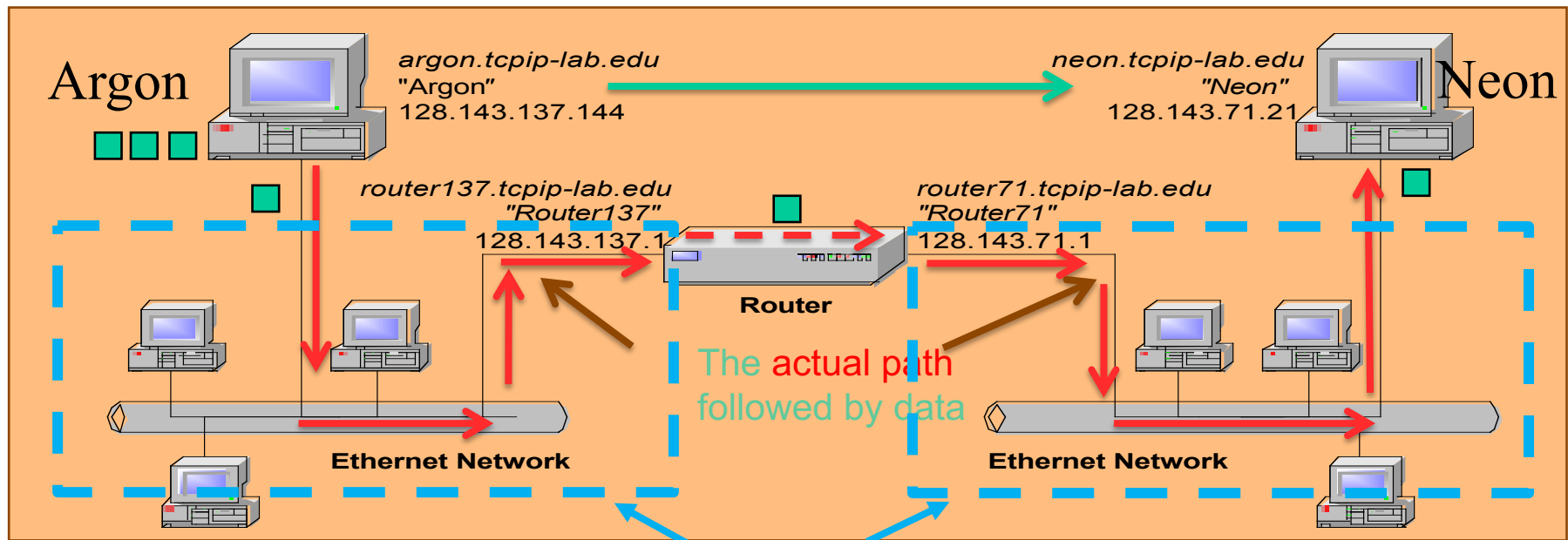
How to determine device MAC addresses

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Question: how to determine a device's MAC address, if all you have is its internet (i.e, IP) address?

Answer: ask for it! Address Resolution Protocol - ARP

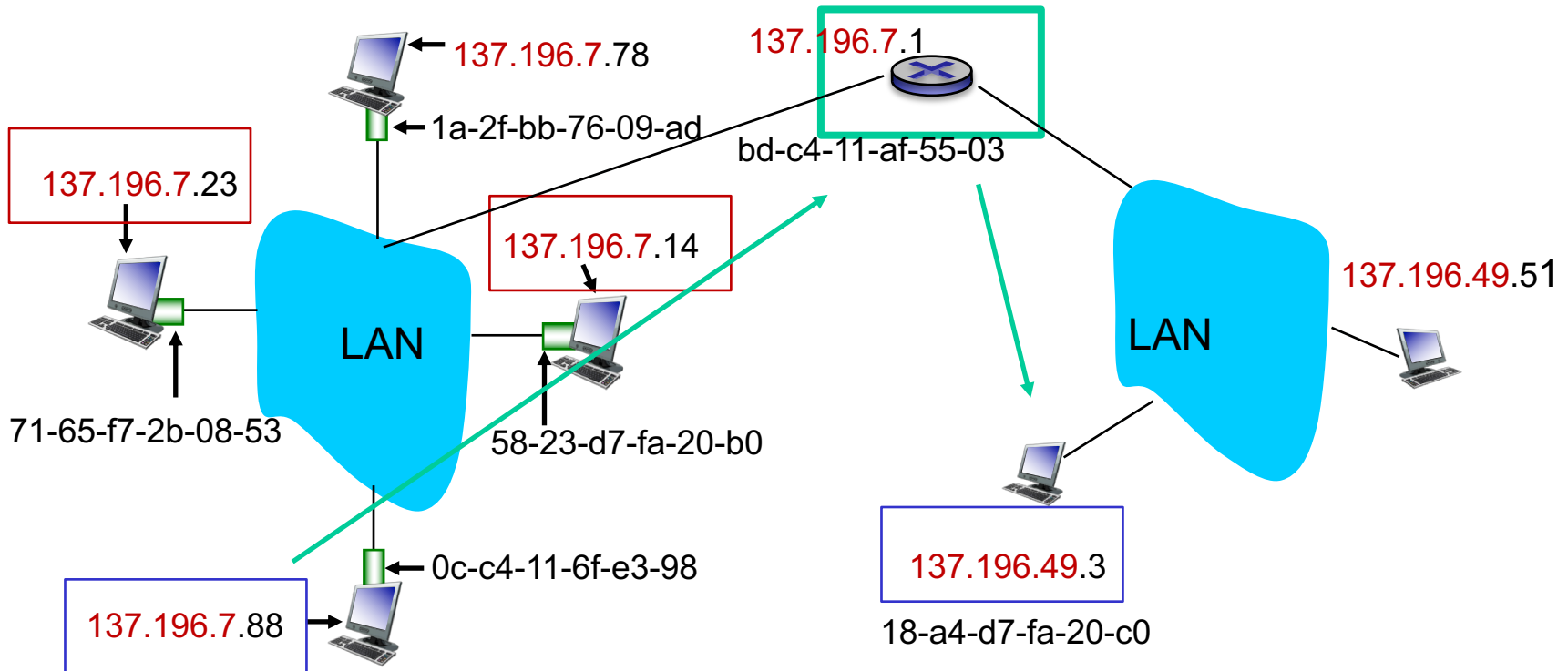
Note: only needed for local/adjacent devices (recall IP hop-by-hop transmission)



Local/Adjacent devices

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All devices are on same “**subnet**”



ARP protocol: same local net

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- A wants to send datagram to B:
 - B has to be on **same local** (physical) network (more later...)
 - B's IP address known to A
 - B's MAC address not known to A
- A **broadcasts** on **local** network **ARP request** packet, containing B's IP address in data field
 - destination MAC address of data link frame = **ff:ff:ff:ff:ff:ff**
- all nodes on local network receive ARP request and read
- B recognizes its IP address in request
- B then sends an **ARP reply** to A with its (B's) MAC address in the data field
 - destination MAC address of data link frame = A's MAC address (B extracted A's MAC address from received ARP request).

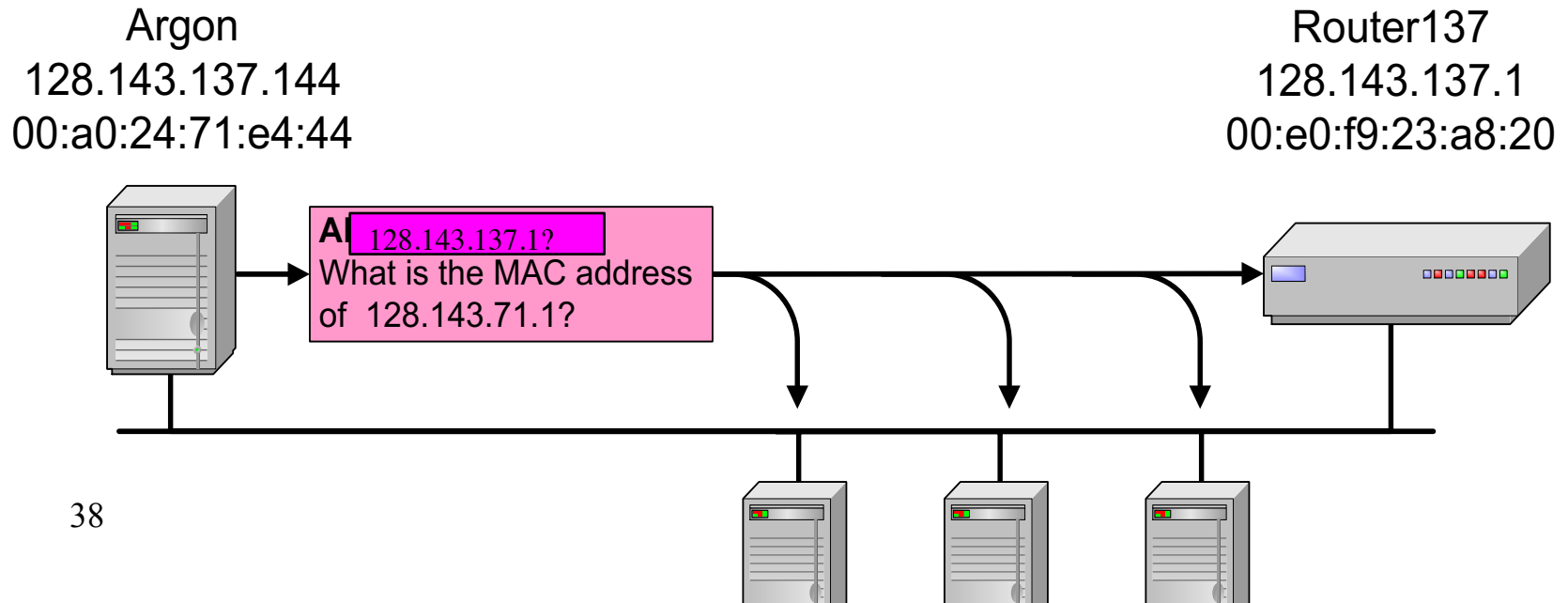
Address Translation with ARP

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ARP Request:

Argon **broadcasts** on data link layer (address: **ff:ff:ff:ff:ff:ff**) an ARP request to all stations on the network:

“What is the hardware address of 128.143.137.1?”

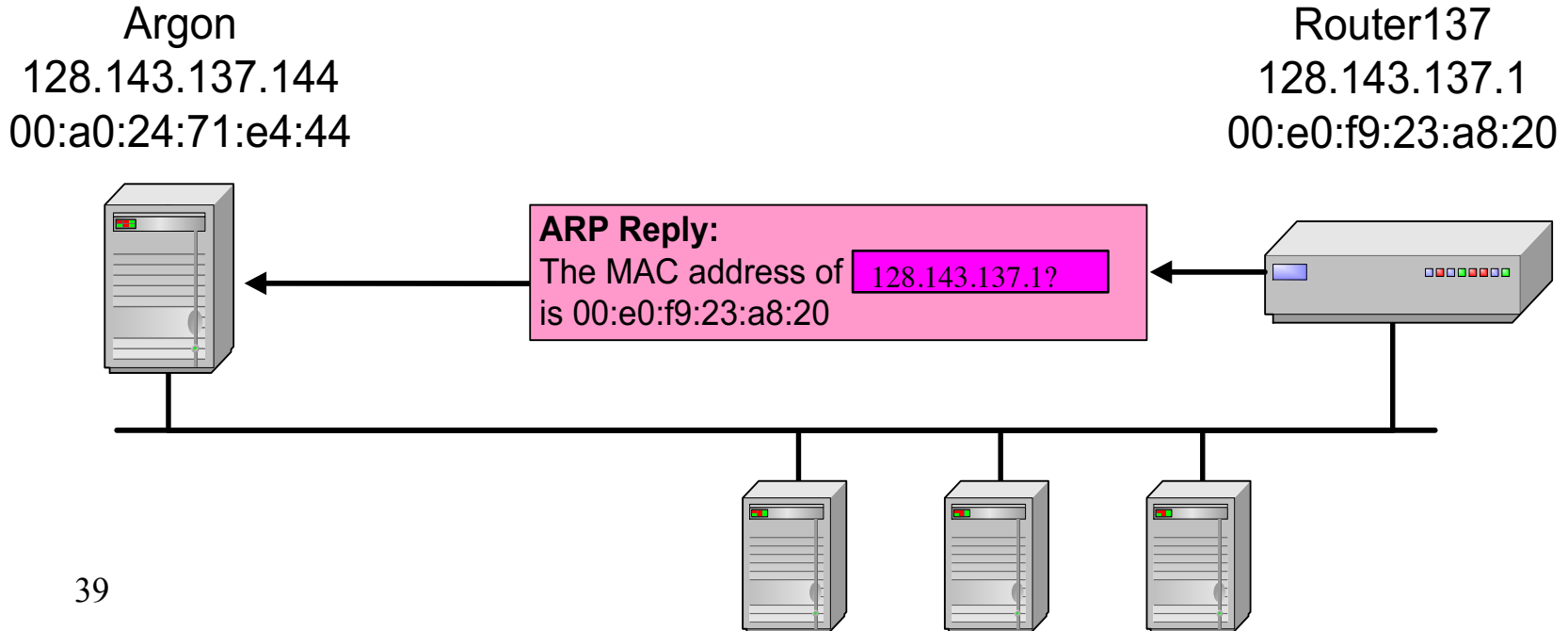


Address Translation with ARP

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ARP Reply:

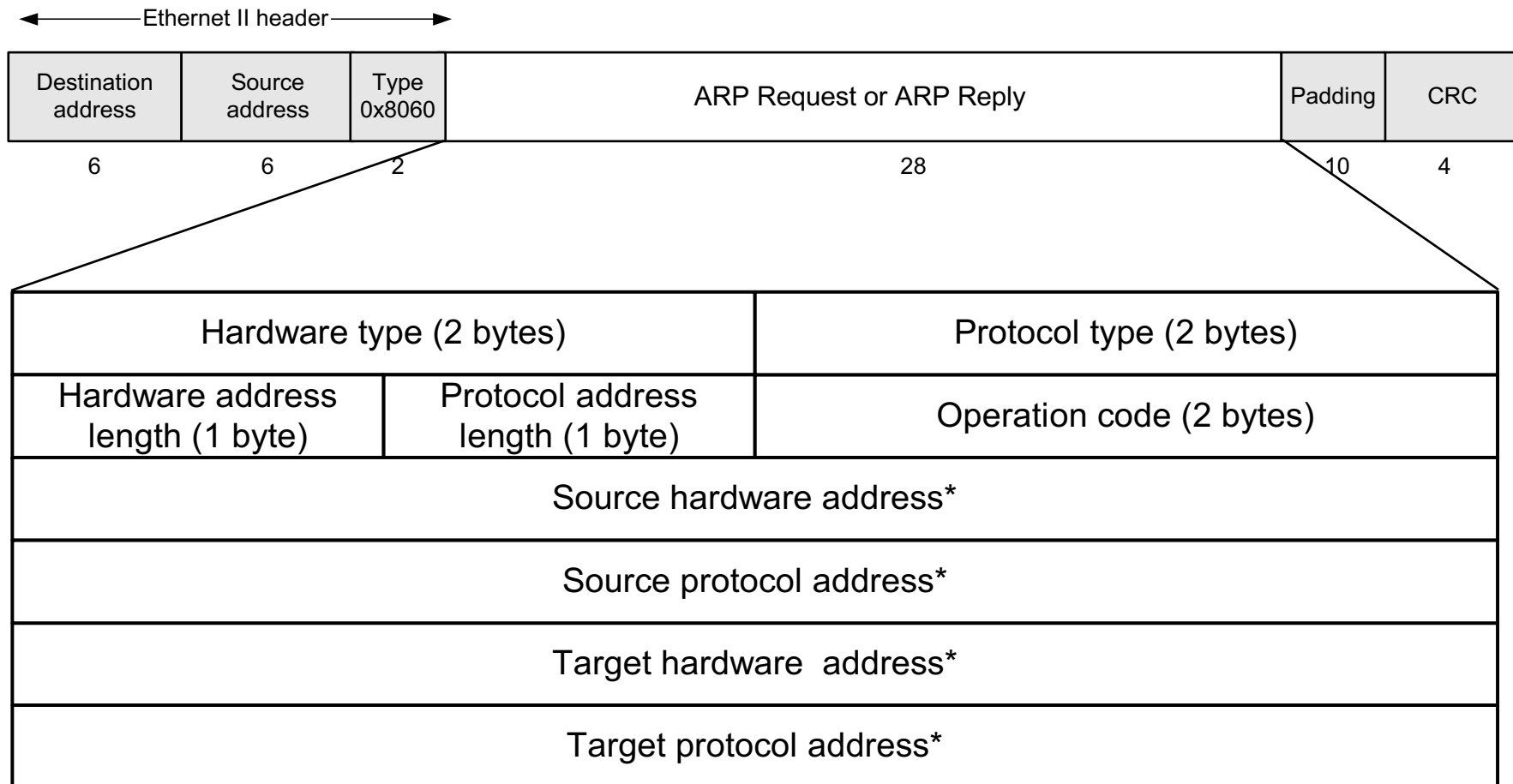
Router 137 responds with an ARP Reply which contains its hardware (MAC) address 00:e0:f9:23:a8:20 and is sent in data link frame to ARGON at address 00:a0:24:71:e4:44



39 of 46 ARP Transport

- ARP message travels in **data portion** of data link layer frame
- we say ARP message is *encapsulated in data link frame*
 - *query/request has:*
 - *destination address: ff:ff:ff:ff:ff:ff*
 - *source address: device's own MAC address*
 - *reply has:*
 - *destination address: of requesting device*
 - *source address: device's own MAC address*
- data portion **padded** with zeroes if ARP message is shorter than **minimum** data link frame (depends on hardware address length and network address length)
- for **Ethernet**: frame type field 0x0806 used for ARP

40 of 46 ARP Packet on Ethernet Link



* Note: The length of the address fields is determined by the corresponding address length fields

Example content of ARP packet

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■ *ARP Request from Argon:*

Source hardware address:	00:a0:24:71:e4:44
Source protocol address:	128.143.137.144
Target hardware address:	00:00:00:00:00:00
Target protocol address:	128.143.137.1

■ *ARP Reply from Router 137:*

Source hardware address:	00:e0:f9:23:a8:20
Source protocol address:	128.143.137.1
Target hardware address:	00:a0:24:71:e4:44
Target protocol address:	128.143.137.144

→ **Result: an entry in ARP cache:**

(128.143.137.1) at 00:e0:f9:23:a8:20 [ether] on eth0

42.0146 ARP Packet Format

- general: can be used with
 - arbitrary hardware address (not just Ethernet)
 - arbitrary protocol address (not just IP)
- variable length address fields (depends on type of datalink protocol)

Retention of Bindings

- sending an ARP request/reply for each IP datagram to same IP address is **inefficient**
- solution -> maintain a table of bindings
 - devices maintain a **cache** of currently used IP addresses and their corresponding hardware addresses
- effect
 - use ARP for first datagram to a destination, place results in table, then quick lookup for MAC address (subsequent packets sent to that same IP address)

ARP Cache

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- A caches (saves) B's IP-MAC address pair in an **ARP table** until information becomes old (times out)

ARP table: < IP address; MAC address; TTL >

- TTL (Time To Live): time after which address mapping will be forgotten (used to be 20 mins, now with mobile devices it is barely 30secs)
 - known as **soft state**: information that times out (goes away) unless refreshed
 - nodes sometimes will send a query to check if remote node is still “valid” - if a response is received, it renews the TTL
- B should cache A's information too, sometimes it needs a “data” request from B to A to enable cache
 - ARP is “plug-and-play”:
 - nodes create their ARP tables *without intervention from net administrator*

Example ARP Cache (table)

(128.143.71.37) at 00:10:4B:C5:D1:15 [ether] on eth0
(128.143.71.36) at 00:B0:D0:E1:17:D5 [ether] on eth0
(128.143.71.35) at 00:B0:D0:DE:70:E6 [ether] on eth0
(128.143.136.90) at 00:05:3C:06:27:35 [ether] on eth1
(128.143.71.34) at 00:B0:D0:E1:17:DB [ether] on eth0
(128.143.71.33) at 00:B0:D0:E1:17:DF [ether] on eth0

Things to know about ARP

- what happens if an ARP request is made for a non-existing host?
 - several ARP requests are made with increasing time intervals between requests.
 - eventually ARP gives up.
- what if a host sends an ARP request for its own IP address?
 - know as gratuitous ARP
 - no response is received hopefully.....
 - this is useful for detecting if there is another device on the local network with the same Internet address (can happen by mistake)